Parameter	Type of Light Bulb			
	Incandescent	Fluorescent	White LED	
			Circa 2010	Circa 2025
Luminous Efficacy (lumens/W)	~12	~40	~70	~150
Useful Lifetime (hours)	~1000	~20,000	~60,000	~100,000
Purchase Price	~$1.50	~$5	~$10	~$5
Estimated Cost over 10 Years	~$410	~$110	~$100	~$40

Figure TF1-7 Even though the initial purchase price of a white LED is several times greater than that of the incandescent light bulb, the total 10-year cost of using the LED is only one-fourth of the incandescent's (in 2010) and is expected to decrease to one-tenth by 2025.

LED照明：原书性能与成本比较

图TF1-7 灯具性能与10年使用成本

参数	白炽灯	荧光灯	白光LED 约2010年	白光LED 原书对约2025年的预测
光效（lm/W）	约12	约40	约70	约150
使用寿命（小时）	约1,000	约20,000	约60,000	约100,000
购买价格（美元）	约1.50	约5	约10	约5
10年估算成本（美元）	约410	约110	约100	约40

尽管白光LED的购买价格比白炽灯高数倍，原书估计其10年总使用成本在2010年前后仅为白炽灯的约四分之一，并预测到2025年前后降至约十分之一。

资料年代说明：表中“约2025年”是原书写作时的预测，不是对2025年实际市场的测量结果。本页只校对、翻译原书数据，不将历史预测改写成已实现的事实。

Concept Question 1-4: What do the electrical permittivity and magnetic permeability of a material account for?

Concept Question 1-5: What are the three branches and associated conditions of electromagnetics?

Exercise 1-1: Given charges $q_1 = 10$ mC, $q_2 = -10$ mC, and $q_3 = 5$ mC, all in free space, what is the direction of the force acting on charge q_3?

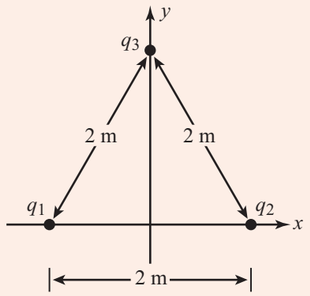


Figure E1.1

Answer: Along $+\hat{\mathbf{x}}$ direction. [See EM (the "EM" symbol refers to the book website: em8e.eecs.umich.edu).]

Exercise 1-2: Two parallel, very long, wires carry currents I_1 and I_2. The magnetic field due to current I_1 alone is $\mathbf{B}_1$. What is the magnetic field due to both currents at a point midway between the wires if

(a) $I_1 = I_2$ and both currents flow along the $+\hat{\mathbf{y}}$ direction?

(b) $I_1 = I_2$, but I_2 flows along the $-\hat{\mathbf{y}}$ direction?

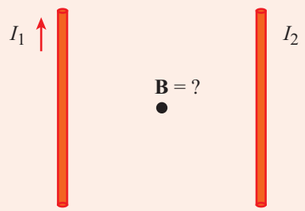


Figure E1.2

Answer: (a) $\mathbf{B} = 0$, (b) $\mathbf{B} = 2\mathbf{B}_1$. (See EM.)

1-4 Traveling Waves

Waves are a natural consequence of many physical processes: Waves manifest themselves as ripples on the surfaces of oceans and lakes; sound waves constitute pressure disturbances that travel through air; mechanical waves modulate stretched strings; and electromagnetic waves carry electric and magnetic fields through free space and material media as microwaves, light, and X-rays. All of these various types of waves exhibit a number of common properties, including:

- ***Moving waves carry energy.***
- ***Waves have velocity;*** it takes time for a wave to travel from one point to another. Electromagnetic waves in a vacuum travel at a speed of 3×10^8 m/s, and sound waves in air travel at a speed approximately a million times slower, specifically 330 m/s. Sound waves cannot travel in a vacuum.
- ***Many waves exhibit a property called linearity.*** Waves that do not affect the passage of other waves are called ***linear*** because they can pass right through each other. The total of two linear waves is simply the sum of the two waves as they would exist separately. Electromagnetic waves are linear, as are sound waves. When two people speak to one another, the sound waves they generate do not interact with one another but simply pass through each other. Water waves are approximately linear; the expanding circles of ripples caused by two pebbles thrown into two locations on a lake's surface do not affect each other. Although the interaction of the two circles may exhibit a complicated pattern, it is simply the linear superposition of two independent expanding circles.

Waves are of two types: ***transient waves*** caused by sudden disturbances and ***continuous periodic waves*** generated by a repetitive source. We encounter both types of waves in this book, but most of our discussion deals with the propagation of continuous waves that vary sinusoidally with time.

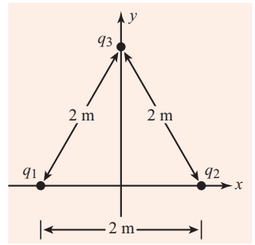
An essential feature of a propagating wave is that it is a self-sustaining disturbance of the medium through which it travels. If this disturbance varies as a function of one space variable, such as the vertical displacement of the string shown in **Fig. 1-10**, we call the wave ***one-dimensional***. The vertical displacement varies with time and with the location along the length of the string. Even though the string rises up into a second dimension, the wave is only one-dimensional

概念题、练习与1-4行波

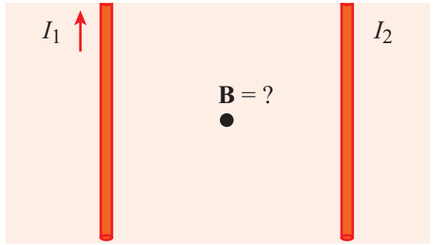
概念题1-4：介质的介电常数与磁导率分别表征什么性质？
概念题1-5：电磁学的三个分支及各自对应的条件是什么？

练习1-1：自由空间中三个电荷的位置见图E1.1，q_1 = 10 mC、q_2 = −10 mC、q_3 = 5 mC。q_3所受合力指向何方？**答案：**沿+x方向。

练习1-2：两条平行、足够长的直导线分别通有电流I_1和I_2。在两导线中间的观察点，仅I_1产生的磁场为$\mathbf{B}_1$。若（a）$I_1 = I_2$，两者均沿+y方向；（b）$I_1 = I_2$，但I_2沿−y方向，求总磁场。**答案：**（a）$\mathbf{B} = 0$；（b）$\mathbf{B} = 2\mathbf{B}_1$。



图E1.1 电荷的位置及间距。



图E1.2 平行载流导线。

原书EM符号指配套网站em8e.eecs.umich.edu。练习1-1的答案由库仑力的对称性核对；练习1-2由右手定则与磁场叠加核对。

1-4 行波

许多物理过程都会产生波：海洋、湖面的涟漪是水波；声波是在空气中传播的压力扰动；绷紧的弦可传播机械波；微波、光和X射线则是携带电场与磁场、能够在自由空间和物质介质中传播的电磁波。这些波具有若干共同性质：

能量传输：行波携带能量。
有限传播速度：波由一点到另一点需要时间。真空中的电磁波速度约为3 × 10^8 m/s；原书取空气声速约330 m/s，比光速低约六个数量级。声波不能在真空中传播。
线性叠加：在线性条件下，两列波可相互穿过，总扰动等于两列波分别存在时扰动之和。日常对话中的声波、两个小石子激起的小振幅水面涟漪，可近似如此处理；干涉图样复杂并不意味着两列波的传播规律被改变。

波可分为突然扰动产生的瞬态波，以及周期性源产生的连续周期波。本书涉及两者，但主要讨论随时间作正弦变化的连续波。传播中的波是能够延续传播的扰动。若扰动只依赖一个空间坐标，即为一维波。例如弦的横向位移虽画在二维平面中，其数值仅随沿弦坐标及时间变化，因此仍是一维波。（接下页）

适用条件校注：线性叠加以介质及波动方程可作线性描述为前提；强声波或非线性介质中的电磁波不能无条件套用。电磁波在真空中传播不需要物质介质。

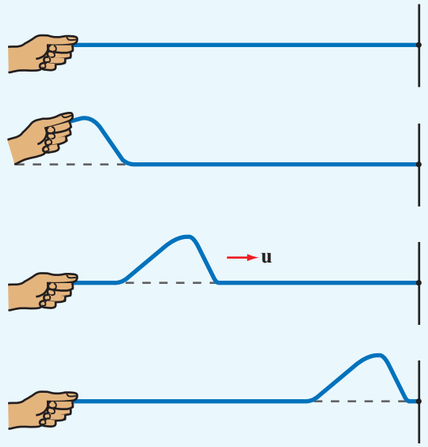


Figure 1-10 A one-dimensional wave traveling on a string.

because the disturbance varies with only one space variable. A ***two-dimensional wave*** propagates out across a surface, like the ripples on a pond (**Fig. 1-11(a)**), and its disturbance can be described by two space variables. By extension, a ***three-dimensional wave*** propagates through a volume, and its disturbance may be a function of all three space variables. Three-dimensional waves may take on many different shapes; they include ***plane waves***, ***cylindrical waves***, and ***spherical waves***. A plane wave is characterized by a disturbance that at a given point in time has uniform properties across an infinite plane perpendicular to its direction of propagation (**Fig. 1-11(b)**). Similarly, for cylindrical and spherical waves, the disturbances are uniform across cylindrical and spherical surfaces (**Figs. 1-11(b)** and **(c)**).

In the material that follows, we examine some of the basic properties of waves by developing mathematical formulations that describe their functional dependence on time and space variables. To keep the presentation simple, we limit our discussion to sinusoidally varying waves whose disturbances are functions of only one space variable, and we defer the discussion of more complicated waves to later chapters.

1-4.1 Sinusoidal Waves in a Lossless Medium

Regardless of the mechanism responsible for generating them, all linear waves can be described mathematically in common terms.

► A medium is said to be ***lossless*** if it does not attenuate the amplitude of the wave traveling within it or on its surface. ◄

By way of an example, let us consider a wave traveling on a lake's surface, and let us assume for the time being that frictional forces can be ignored, thereby allowing a wave generated on the water's surface to travel indefinitely with no loss in energy. If y denotes the height of the water's surface

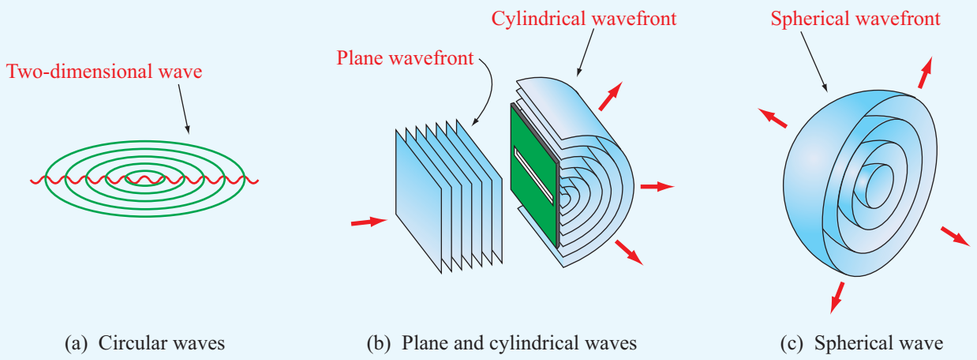


Figure 1-11 Examples of two-dimensional and three-dimensional waves: (a) circular waves on a pond, (b) a plane light wave exciting a cylindrical light wave through the use of a long narrow slit in an opaque screen, and (c) a sliced section of a spherical wave.

波的维数与无损介质中的正弦波

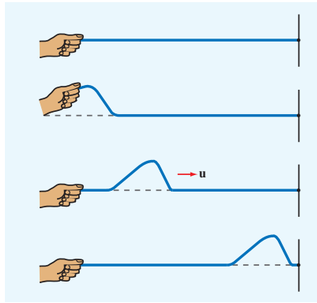


图1-10 沿弦传播的一维波，u为传播速度。

（承上页）一维波的扰动仅依赖一个空间坐标。二维波在表面上传播，例如池塘涟漪，需两个空间坐标描述。三维波在空间体积中传播，一般可依赖三个空间坐标，其形式包括平面波、柱面波和球面波。

在某一时刻，理想平面波在垂直于传播方向的整个平面上具有相同的场分布。原书同样用柱面、球面上的相同扰动说明理想柱面波和球面波，见图1-11。以下将用数学表达式描述波对时间与空间的依赖，先限于随时间作正弦变化、并仅依赖一个空间变量的情形；更复杂的波留到后续章节。

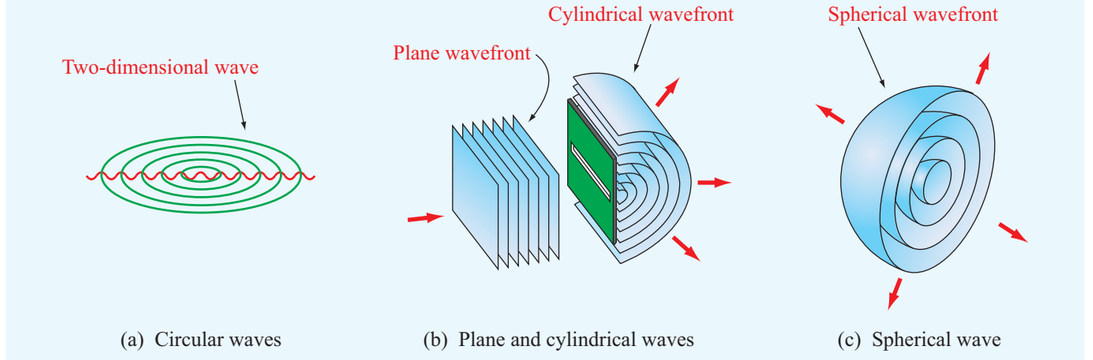


图1-11 （a）池塘中的圆形波；（b）平面光波穿过不透明屏上的狭长缝隙，形成柱面光波；（c）球面波的剖切示意。

图中Two-dimensional wave：二维波；Plane、Cylindrical、Spherical wavefront分别为平面、柱面、球面波前。波前形状与场矢量的方向是不同概念。

1-4.1 无损介质中的正弦波

无论产生机制如何，线性波都可用共同的数学形式描述。原书把无损介质描述为不使其中或其表面传播的波发生振幅衰减的介质。以湖面波为例，暂时忽略摩擦，便可研究不因耗散而损失能量的波。设y表示水面高度。（接下页）

定义校注：无损指没有介质耗散造成的能量损失。即使介质无损，球面波或柱面波仍可因几何扩散而随距离减小振幅；原书“不衰减振幅”的说法须限于排除了这类几何因素的情形。

relative to the mean height (undisturbed condition) and x denotes the distance of wave travel, the functional dependence of y on time t and the spatial coordinate x has the general form

$$y(x,t) = A\cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0\right) \text{ (m)}, \tag{1.17}$$

where A is the ***amplitude*** of the wave, T is its ***time period***, λ is its ***spatial wavelength***, and ϕ_0 is a ***reference phase***. The quantity $y(x,t)$ also can be expressed in the form

$$y(x,t) = A\cos\phi(x,t) \quad \text{(m)}, \tag{1.18}$$

where

$$\phi(x,t) = \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0\right) \quad \text{(rad)}. \tag{1.19}$$

The angle $\phi(x,t)$ is called the ***phase*** of the wave, and it should not be confused with the reference phase ϕ_0, which is constant with respect to both time and space. Phase is measured by the same units as angles, that is, radians (rad) or degrees, with 2π radians $= 360°$.

Let us first analyze the simple case when $\phi_0 = 0$:

$$y(x,t) = A\cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right) \quad \text{(m)}. \tag{1.20}$$

The plots in **Fig. 1-12** show the variation of $y(x,t)$ with x at $t = 0$ and with t at $x = 0$. The wave pattern repeats itself at a spatial period λ along x and at a temporal period T along t.

If we take time snapshots of the water's surface, the height profile $y(x,t)$ would exhibit the sinusoidal patterns shown in **Fig. 1-13**. All three profiles correspond to three different values of t, and the spacing between peaks is equal to the wavelength λ, even though the patterns are shifted relative to one another because they correspond to different observation times. Because the pattern advances along the $+x$ direction at progressively increasing values of t, $y(x,t)$ is called a wave traveling in the $+x$ direction. If we track a given point on the wave, such as the peak P, and follow it in time, we can measure the ***phase velocity*** of the wave. At the peaks of the wave pattern, the phase $\phi(x,t)$ is equal to zero or multiples of 2π radians. Thus,

$$\phi(x,t) = \frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = 2n\pi, \quad n = 0,1,2,\ \ldots \tag{1.21}$$

Had we chosen any other fixed height of the wave, say y_0, and monitored its movement as a function of t and x, this again would have been equivalent to setting the phase $\phi(x,t)$ constant such that

$$y(x,t) = y_0 = A\cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right) \tag{1.22}$$

or

$$\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = \cos^{-1}\left(\frac{y_0}{A}\right) = \text{ constant}. \tag{1.23}$$

The apparent velocity of that fixed height is obtained by taking the time derivative of Eq. (1.23), so

$$\frac{2\pi}{T} - \frac{2\pi}{\lambda}\frac{dx}{dt} = 0, \tag{1.24}$$

which gives the ***phase velocity*** u_p as

$$u_p = \frac{dx}{dt} = \frac{\lambda}{T} \quad \text{(m/s)}. \tag{1.25}$$

▶ The phase velocity, also called the ***propagation velocity***, is ***the velocity of the wave pattern*** as it moves across the water's surface. ◀

The water itself mostly moves up and down; when the wave moves from one point to another, the water does not move physically along with it.

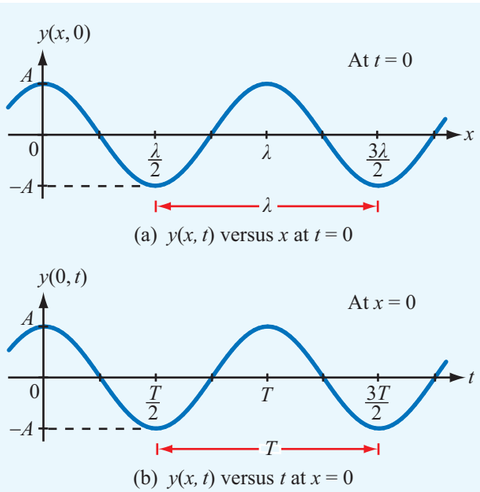


Figure 1-12 Plots of $y(x,t) = A\cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right)$ as a function of (a) x at $t = 0$ and (b) t at $x = 0$.

正弦行波的相位与相速度

（承上页）设y为水面相对未扰动平均位置的高度，x为沿传播方向的坐标，则沿+x方向传播的无损正弦波可写为：

$$y(x,t) = A\cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0\right) \text{ (m)}, \tag{1.17}$$

A为振幅，T为时间周期，λ为波长，φ_0为不随时间和空间变化的参考相位。也可写成式（1.18），其中相位φ(x,t)由式（1.19）给出：

$$y(x,t) = A\cos\phi(x,t) \quad \text{(m)}, \tag{1.18}$$

$$\phi(x,t) = \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0\right) \quad \text{(rad)}. \tag{1.19}$$

相位φ(x,t)随x、t变化，不应与常数φ_0混淆。角度可用弧度或度表示，2π rad = 360°。令φ_0 = 0，得到：

$$y(x,t) = A\cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right) \quad \text{(m)}. \tag{1.20}$$

图1-12分别给出t = 0时随x变化的空间波形，以及x = 0时随t变化的时间波形；两者的周期分别为λ、T。图1-13中的三个时刻虽然波峰位置不同，相邻波峰间距却都为λ。随着时间增大，波形向+x移动，故称沿+x传播的行波。

图1-12见左侧原书：（a）固定t = 0，观察y(x,0)；（b）固定x = 0，观察y(0,t)。横轴分别为位置x与时间t，纵轴均为位移。

跟踪同一波峰P或其他固定相位点，可求得相速度。波峰满足相位为2π的整数倍：

$$\phi(x,t) = \frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = 2n\pi, \quad n = 0,1,2,\ \ldots \tag{1.21}$$

同样，对任意选定高度y_0，沿同一相位分支连续跟踪，便可把相位设为常数：

$$y(x,t) = y_0 = A\cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right) \tag{1.22}$$

$$\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = \cos^{-1}\left(\frac{y_0}{A}\right) = \text{ constant}. \tag{1.23}$$

沿该固定相位点的轨迹对时间求导，即得式（1.24）和相速度式（1.25）：

$$\frac{2\pi}{T} - \frac{2\pi}{\lambda}\frac{dx}{dt} = 0, \tag{1.24}$$

$$u_p = \frac{dx}{dt} = \frac{\lambda}{T} \quad \text{(m/s)}. \tag{1.25}$$

相速度是波形或等相位点的传播速度。水质点主要在平衡位置附近振动，并不随波形整体向前迁移。

数学校注：一般波峰条件应取n为任意整数；固定y_0可对应多个相位值，推导时应固定其中一个连续分支，不能在不同波峰或不同反余弦分支间跳换。

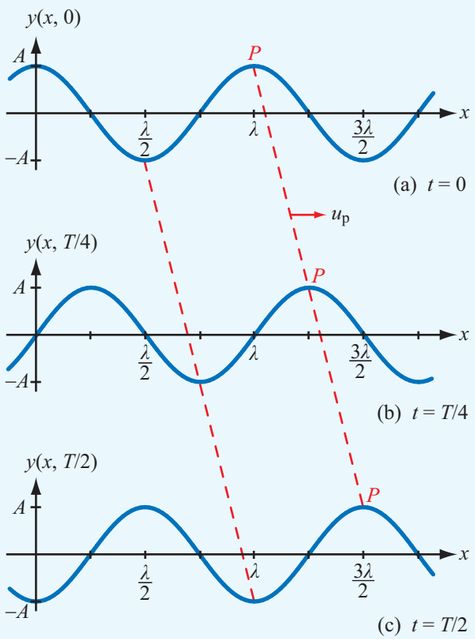


Figure 1-13 Plots of $y(x,t) = A\cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right)$ as a function of x at (a) $t = 0$, (b) $t = T/4$, and (c) $t = T/2$. Note that the wave moves in the $+x$ direction with a velocity $u_p = \lambda/T$.

The ***frequency*** of a sinusoidal wave, f, is the reciprocal of its time period T:

$$f = \frac{1}{T} \quad \text{(Hz)}. \tag{1.26}$$

Combining the preceding two equations yields

$$u_p = f\lambda \quad \text{(m/s)}. \tag{1.27}$$

The wave frequency f is measured in cycles per second and has been assigned the unit hertz (Hz), named in honor of the German physicist Heinrich Hertz (1857–1894), who pioneered the development of radio-wave instrumentation.

Using Eq. (1.26), Eq. (1.20) can be rewritten in a more compact form as

$$y(x,t) = A\cos\left(2\pi f t - \frac{2\pi}{\lambda}x\right) = A\cos(\omega t - \beta x), \tag{1.28}$$

(wave moving along $+x$ direction)

where ω is the ***angular velocity*** of the wave and β is its ***phase constant*** (or ***wavenumber***), defined as

$$\omega = 2\pi f \quad \text{(rad/s)}, \tag{1.29a}$$

$$\beta = \frac{2\pi}{\lambda} \quad \text{(rad/m)}. \tag{1.29b}$$

In terms of these two quantities,

$$u_p = f\lambda = \frac{\omega}{\beta}. \tag{1.30}$$

So far, we have examined the behavior of a wave traveling in the $+x$ direction. To describe a wave traveling in the $-x$ direction, we reverse the sign of x in Eq. (1.28):

$$y(x,t) = A\cos(\omega t + \beta x). \tag{1.31}$$

(wave moving along $-x$ direction)

▶ The direction of wave propagation is easily determined by inspecting the signs of the t and x terms in the expression for the phase $\phi(x,t)$ given by Eq. (1.19): If one of the signs is positive and the other is negative, then the wave is traveling in the positive x direction, and if both signs are positive or both are negative, then the wave is traveling in the negative x direction. The constant phase reference ϕ_0 has no influence on either the speed or the direction of wave propagation. ◀

We now examine the role of the phase reference ϕ_0 given previously in Eq. (1.17). If ϕ_0 is not zero, then Eq. (1.28) should be written as

$$y(x,t) = A\cos(\omega t - \beta x + \phi_0). \tag{1.32}$$

A plot of $y(x,t)$ as a function of x at a specified t or as a function of t at a specified x is shifted in space or time, respectively, relative to a plot with $\phi_0 = 0$ by an amount

频率、波数与传播方向

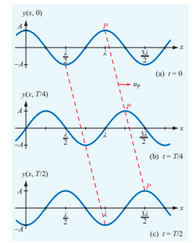


图1-13 t = 0、T/4、T/2时的波形；同一波峰P以$u_p = \lambda/T$沿+x方向移动。

正弦波频率f是周期T的倒数。结合相速度表达式可得：

$$f = \frac{1}{T} \quad \text{(Hz)}. \tag{1.26}$$

$$u_p = f\lambda \quad \text{(m/s)}. \tag{1.27}$$

频率表示每秒重复的周期数，单位为赫兹（Hz），得名于无线电波实验先驱Heinrich Hertz（1857—1894）。利用式（1.26），式（1.20）可写为：

$$y(x,t) = A\cos\left(2\pi f t - \frac{2\pi}{\lambda}x\right) = A\cos(\omega t - \beta x), \tag{1.28}$$

(wave moving along $+x$ direction)

ω称为角频率，β称为相位常数或波数，分别定义为：

$$\omega = 2\pi f \quad \text{(rad/s)}, \tag{1.29a}$$

$$\beta = \frac{2\pi}{\lambda} \quad \text{(rad/m)}. \tag{1.29b}$$

用这两个量表示相速度，有：

$$u_p = f\lambda = \frac{\omega}{\beta}. \tag{1.30}$$

要描述沿−x方向传播的波，只须将式（1.28）中x项的符号反转：

$$y(x,t) = A\cos(\omega t + \beta x). \tag{1.31}$$

(wave moving along $-x$ direction)

观察式（1.19）相位中t项和x项的符号即可判断传播方向：在ω、β取正值的约定下，两项异号表示沿+x传播，同号表示沿−x传播。常数参考相位φ_0不改变传播方向或相速度。

式（1.17）的参考相位φ_0不为零时，沿+x方向传播的波为：

$$y(x,t) = A\cos(\omega t - \beta x + \phi_0). \tag{1.32}$$

固定时刻观察空间波形，或固定位置观察时间波形，相对于$\varphi_0 = 0$的参考波都会发生与φ_0成正比的平移。（接下页）